

POCKET-SIZE READING LAMP

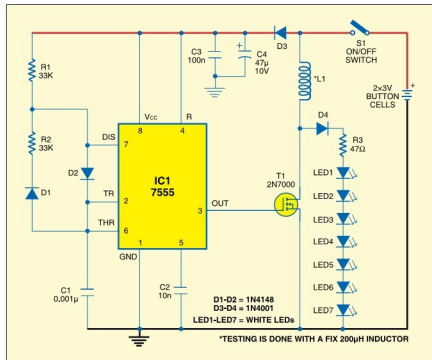


■ T.A. BABU

This mini reading light combines the advantages of lithium button cells and white LEDs. While white LEDs are super-bright, lithium cells are small in size and last long.

Generating high voltage from a low-voltage power supply basically requires an oscillator. Timer IC 555 is configured here as an astable multivibrator with 'on' time of approximately 22 micro-seconds.

When you press switch S1, IC1 receives 6V through diode D3 and the timer oscillates to make MOSFET T1 conduct and cut-off alternately with a switching frequency of 22 kHz. When the MOSFET conducts, the current through inductor L1 increases gradually and it stores energy. When the MOSFET cuts off, the



energy stored in the inductor drives the LEDs. This flyback voltage is sufficient to make the string of seven LEDs glow.

Assemble the circuit on a general-purpose PCB and enclose in a small box. Connect on/off switch on top of the box. ●

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